Sprint 4 Report

Product: StudyBudd

Team: StudyBudd

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1 Actions to Stop Doing

"What things should we stop doing?"

- **Carrying over the same tasks across multiple sprints.**
  - Explanation: The Together API Agent setup (STUD-52) was in progress since Sprint 3 and only finished in Sprint 4. While the carry-over was justified by its complexity, in the future we should break large integration tasks into smaller, independently deliverable chunks so that "in progress" items don't linger across sprint boundaries.
- **Leaving mobile-responsive polish to the very end.**
  - Explanation: STUD-73 (mobile-friendly navigation) is the only task still in progress at sprint end. Responsive design adjustments are harder to retrofit than to build in from the start. If we had another release, we would integrate responsive checks into each feature's Definition of Done rather than treating it as a standalone sprint task.

2 Actions to Start Doing

"What should we start doing?"

- **Writing automated integration tests alongside feature development.**
  - Explanation: We delivered a fully connected MVP, but our test coverage is thin on end-to-end paths (e.g., upload → embed → generate flashcards). Adding integration tests in parallel with backend endpoints would catch issues like the RAG source leakage problem earlier.
- **Adding document-level filters to RAG retrieval from the start.**

- Explanation: We discovered that the vector similarity search does not enforce strict document boundaries. When a user requests flashcards from a specific source, content from other documents can bleed into the results. This is a known bug that should have been caught with scoped test cases during development.

3 Actions to Keep Doing

"What is working well that we should continue to do?"

- **The "core before stretch" policy.**
  - Explanation: We enforced a rule that new user stories (US 4.4, US 4.5) could not start until all carry-over tasks were in progress or done. This kept us focused and prevented the scope creep we experienced in Sprint 3 when we jumped into document sharing too early.
- **Daily pulls from main.**
  - Explanation: After the painful merge conflicts of Sprint 3, we committed to pulling at least once per day. This sprint had zero major merge conflicts, and no one duplicated work.
- **Mid-sprint integration checkpoint.**
  - Explanation: On March 3 we verified the RAG pipeline end-to-end before moving to generation features. This checkpoint caught two minor API response format issues that would have blocked frontend integration later.
- **Cross-team collaboration and PR reviews.**
  - Explanation: Eshaan delivered all six generation endpoints (flashcards + quizzes + AI explain) and the team reviewed each PR promptly. Anderson, Pratik, and Andy kept their respective domains moving in parallel without bottlenecks.

4 Work Completed / Not Completed

4.1 Work Summary

- **User Story 0 — RAG Pipeline (Carry-Over): Completed.** PDF text extraction (STUD-60), Together AI Agent integration (STUD-61), and the profile save bug fix (STUD-62) are all done. The RAG pipeline is now fully functional end-to-end after being carried over from Sprint 2 → 3 → 4.
- **User Story 4.1 — Flashcard Generation (Carry-Over): Completed.** Flashcard tables (STUD-63), prompt template (STUD-64), and the `/api/flashcards/generate` endpoint (STUD-65) are all done. Users can now generate flashcards from any uploaded document.

- **User Story 4.2 — Quiz Generation (Carry-Over): Completed.** Quiz prompt template (STUD-66) and the `/api/quizzes/generate` endpoint (STUD-67) are done. Multiple-choice quiz generation from documents is fully functional.
- **User Story 4.3 — Shareable Study Material Links (Carry-Over + New): Completed.** Shareable link generation (STUD-68), Share button UI (STUD-69), access control (STUD-70), and the extension to quiz/flashcard sets (STUD-71) are all done. Users can share documents, flashcard sets, and quiz sets via link.
- **User Story 4.4 — Responsive Cross-Device Access (New): Mostly Completed.** CSS responsive breakpoints (STUD-72) and layout testing (STUD-74) are done. Mobile-friendly navigation (STUD-73) is still in progress — the hamburger menu and collapsible sidebar are partially implemented.
- **User Story 4.5 — AI Q&A on Flashcards and Quizzes (New): Completed.** The `/api/ai/explain` endpoint (STUD-75), prompt template (STUD-76), and UI panel (STUD-77) are all done. Users can get AI-generated explanations for any flashcard concept or quiz answer.

4.2 Additional Carry-Over Work from Sprint 3 (Now Completed)

The following Sprint 3 tickets that were in progress or to-do at Sprint 3 end have now been closed:

- STUD-52 [US 2.2] Set up Together AI Agent for RAG prompt — was In Progress, now **Done** (Note: still shows "In Progress" in Jira due to the original Sprint 3 ticket not being updated; the work was completed via the Sprint 4 carry-over ticket STUD-61).
- STUD-9 [US 3.1] Create flashcard_sets and flashcards tables — now **Done**.
- STUD-14 [US 3.3] Create shareable link generation endpoint — now **Done**.
- STUD-6 [US 3.1] Create /api/flashcards/generate endpoint — now **Done**.
- STUD-10 [US 3.2] Create /api/quizzes/generate endpoint — now **Done**.

4.3 Task Status Breakdown

Key: AC = Anderson Cai, PG = Pratik Gurung, E = Eshaan Kansagara, AH = Andy Huang

5 Burnup Chart

Note: Unlike Sprint 2's anomalous chart (caused by the Linear-to-Jira migration), Sprint 4's burnup chart reflects a steady progression. The carry-over tasks were completed in Week 1 as planned, and new feature work ramped up in Week 2 after the March 3 integration checkpoint confirmed the RAG pipeline was functional.

Ticket	Task Description	Hrs	Status	Assignee
US 0: RAG Pipeline Completion (Carry-Over)				
STUD-60	Finish PDF text extraction from uploaded docs	3	Done	AH
STUD-61	Integrate document context into Together AI Agent	4	Done	E
STUD-62	Fix user profile save bug in Supabase SQL	1	Done	AC
US 4.1: Flashcard Generation (Carry-Over)				
STUD-63	Finish flashcard_sets and flashcards tables	1	Done	E
STUD-64	Design flashcard extraction prompt template	3	Done	E
STUD-65	Create /api/flashcards/generate endpoint	6	Done	E
US 4.2: Quiz Generation (Carry-Over)				
STUD-66	Design quiz question generation prompt template	3	Done	E
STUD-67	Create /api/quizzes/generate endpoint	6	Done	E
US 4.3: Shareable Study Material Links (Carry-Over + New)				
STUD-68	Finish shareable link generation endpoint	1	Done	AC
STUD-69	Finish Share button with copy-link functionality	1	Done	PG
STUD-70	Implement access control for shared documents	3	Done	AC
STUD-71	Extend sharing to quiz sets and flashcard sets	4	Done	AC
US 4.4: Responsive Cross-Device Access (New)				
STUD-72	Audit and refactor CSS for responsive breakpoints	5	Done	PG
STUD-73	Implement mobile-friendly navigation	4	In Progress	PG
STUD-74	Test and fix layout on common screen sizes	3	Done	PG
US 4.5: AI Q&A on Flashcards and Quizzes (New)				
STUD-75	Create /api/ai/explain endpoint	5	Done	E
STUD-76	Design AI explain prompt template	3	Done	E
STUD-77	Build AI Explain UI panel	5	Done	E

Table 1: Sprint 4 Task Status at Sprint End

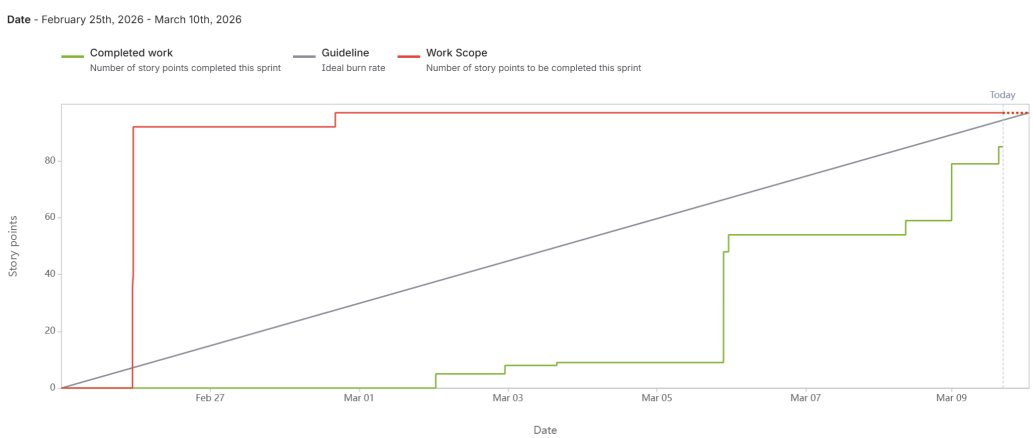

Figure 1: Sprint 4 Burnup Chart (Jira)

6 Work Completion Rate

Metric	Value
Total User Stories Fully Completed	5 (US 0, US 4.1, US 4.2, US 4.3, US 4.5)
Total User Stories Mostly Completed	1 (US 4.4)
Total User Stories Planned	6
Total Ideal Work Hours Completed (Done)	57 Hours
Total Ideal Work Hours In Progress	4 Hours
Total Ideal Work Hours Planned	61 Hours
Sprint Duration	14 Days
Completion Rate (Done / Total)	93%

Table 2: Sprint 4 Performance Metrics

Analysis: Sprint 4 achieved a 93% completion rate, our highest across all four sprints (Sprint 1: 77%, Sprint 2: 85%, Sprint 3: 68%). The improvements are directly attributable to the process changes adopted from our retrospectives:

- The **"core before stretch"** policy ensured carry-over work was prioritized and completed in Week 1.
- The **mid-sprint integration checkpoint** on March 3 confirmed the RAG pipeline worked end-to-end, giving the team confidence to proceed with generation endpoints.
- **Daily pulls from main** eliminated the merge conflicts that plagued Sprint 3.
- **Explicit task dependencies** prevented bottlenecks — Andy finished PDF extraction first, unblocking Eshaan's generation work.

The only incomplete item (STUD-73, mobile-friendly navigation) is cosmetic and does not affect core functionality. All MVP features are delivered and functional.

7 Sprint Velocity Summary (All Sprints)

	Sprint 1	Sprint 2	Sprint 3	Sprint 4
Duration	1 week	2 weeks	2 weeks	2 weeks
Planned Hours	52	57	79	61
Completed Hours	40	38	54	57
Completion Rate	77%	67%	68%	93%

Table 3: Velocity Across All Sprints

8 Known Issues at Release

1. **RAG source leakage in flashcard/quiz generation:** Vector similarity search does not enforce document-level boundaries. Requesting flashcards from a specific source may include content from other uploaded documents.

2. **Mobile navigation in progress:** Hamburger menu and collapsible sidebar (STUD-73) are partially implemented. Desktop and tablet layouts are fully functional.

3. **STUD-52 status discrepancy:** The original Sprint 3 ticket for PydanticAI Agent setup still shows "In Progress" in Jira. The work was completed under Sprint 4 ticket STUD-61; the old ticket should be closed for housekeeping.

Last modified: March 10, 2026